Wazuh SOC Home Lab

Phase 2 Implementation Report

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Introduction:

This lab report is a continuation of the previously published SOC home lab environment built using Wazuh for security monitoring and event analysis. In this phase, the focus moved beyond basic deployment to implementing practical detection and response mechanisms within the lab environment. The setup consisted of a Wazuh server, a Kali Linux machine, and a Windows monitored endpoint, allowing security events to be generated and analyzed in a controlled environment.

During this lab, several defensive monitoring capabilities were implemented. File Integrity Monitoring (FIM) was configured to track critical system directories and detect unauthorized file changes. Custom Wazuh rules were created to identify specific security events such as new user account creation and SSH brute force attempts. In addition, an active response mechanism was configured to automatically block attacking IP addresses after repeated failed SSH login attempts, demonstrating how automated threat response can be integrated into a SOC environment.

Objective:

The objective of this lab was to enhance the SOC home lab environment by implementing detection and automated response capabilities using Wazuh. The focus was on monitoring system activity, detecting suspicious behavior, and responding to potential security threats.

• File Integrity Monitoring (FIM)
• Custom Detection Rules
• Active Response Automation

1) File Integrity Monitoring (FIM):

Overview: File Integrity Monitoring (FIM) is a security mechanism that detects changes made to important files and directories on a system. It helps identify unauthorized modifications, file creation, or deletion that could indicate malicious activity or system compromise.

Implementation (Linux):

- Configured monitoring for sensitive directories such as /etc /var/log.

```
<!-- File integrity monitoring -->
<syscheck>
  <disabled>no</disabled>

  <!-- Frequency that syscheck is executed default every 12 hours -->
  <frequency>43200</frequency>

  <scan_on_start>yes</scan_on_start>

  <!-- Generate alert when new file detected -->
  <alert_new_files>yes</alert_new_files>
                                [ Search Wrapped ]
```

- Created test file.

```
┌──(kali㉿kali)-[/]
└─$ sudo touch /etc/soc_test_file

┌──(kali㉿kali)-[/]
└─$ sudo nano /etc/soc_test_file

┌──(kali㉿kali)-[/]
└─$ sudo rm /etc/soc_test_file

┌──(kali㉿kali)-[/]
└─$
```

- The Wazuh dashboard successfully generated alerts when file changes were performed on the monitored systems.

Mar 13, 2026 @ 00:26:20.616 - Mar 14, 2026 @ 00:26:20.616

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timestamp	agent.name	syscheck.path
Mar 14, 2026 @ 00:25:08.2...	KaliAttacker	/var/log/journal/4635cf0200da4eaea3db9302ed15368d..
Mar 14, 2026 @ 00:25:01.0...	KaliAttacker	/var/log/journal/4635cf0200da4eaea3db9302ed15368d..
Mar 14, 2026 @ 00:24:10.0...	KaliAttacker	/var/log/journal/4635cf0200da4eaea3db9302ed15368d..
Mar 14, 2026 @ 00:23:51.0...	KaliAttacker	/var/log/journal/4635cf0200da4eaea3db9302ed15368d..
Mar 14, 2026 @ 00:23:43.5...	KaliAttacker	/var/log/journal/4635cf0200da4eaea3db9302ed15368d..
Mar 14, 2026 @ 00:23:36.5...	KaliAttacker	/var/log/journal/4635cf0200da4eaea3db9302ed15368d..
Mar 14, 2026 @ 00:23:30.8...	KaliAttacker	/var/log/journal/4635cf0200da4eaea3db9302ed15368d..
Mar 14, 2026 @ 00:23:25.3...	KaliAttacker	/var/log/journal/4635cf0200da4eaea3db9302ed15368d..
Mar 14, 2026 @ 00:19:53.5...	KaliAttacker	/var/log/journal/4635cf0200da4eaea3db9302ed15368d..

Document Details

View surrounding documents | View single document

syscheck.gname_after	systemd-journal
syscheck.inode_after	2368329
syscheck.md5_after	d4fa97bb142fe0b9 84dde1f
syscheck.md5_before	4e2ad7c19e46af76e5d0c4db9 d902b40
syscheck.mode	realtime
syscheck.mtime_after	Mar 14, 2026 @ 00:26:52.0 00
syscheck.path	/var/log/journal/4635cf02 00da4eaea3db9302ed15368d/ system.journal
syscheck.perm_after	rw-r-----
syscheck.sha1_after	357dfb3fc89233a710e21a494 2b37df99abe3306
syscheck.sha1_before	41952b197fa9a0cef29840008 df409fc522eafaa
syscheck.sha256_after	0016bbddf275e843bd368e9e8 eb5c71724bbb4b6e0a54b939b 053383fbdaa27a
syscheck.sha256_before	ef626e3884201054b438aeff1 fd479242afb40eb3fc63981cc 3a12c3f7a25699
syscheck.size_after	8,388,608

Filter for field present

Implementation (Windows):

- Created test file and removed it.

```
Administrator: Windows PowerShell
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> New-Item C:\Windows\System32\soc_test.txt


    Directory: C:\Windows\System32


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----          3/14/2026   5:45 AM              0 soc_test.txt


PS C:\Windows\system32> notepad C:\Windows\System32\soc_test.txt
PS C:\Windows\system32> Remove-Item C:\Windows\System32\soc_test.txt
PS C:\Windows\system32>
```

- The Wazuh dashboard successfully generated alerts when file changes were performed on the monitored systems.

File Integrity M... | Windows10

53 hits

Mar 8, 2026 @ 00:00:00.000 - Mar 14, 2026 @ 23:59:59.999

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timestamp	agent.name	syscheck.path
Mar 14, 2026 @ 09:25:30.9...	Windows10	c:\users\user\downloads\test\test1.txt.txt
Mar 14, 2026 @ 09:25:27.5...	Windows10	c:\users\user\downloads\test\test1.txt.txt
Mar 14, 2026 @ 09:25:27.1...	Windows10	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...
Mar 14, 2026 @ 09:25:27.1...	Windows10	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...
Mar 14, 2026 @ 09:25:27.0...	Windows10	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...
Mar 14, 2026 @ 09:25:26.6...	Windows10	c:\users\user\downloads\test\new text document.txt
Mar 14, 2026 @ 09:25:26.5...	Windows10	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...
Mar 14, 2026 @ 09:25:26.5...	Windows10	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...
Mar 14, 2026 @ 09:25:26.4...	Windows10	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...
Mar 14, 2026 @ 09:25:26.4...	Windows10	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...
Mar 14, 2026 @ 09:25:26.4...	Windows10	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...
Mar 14, 2026 @ 09:25:26.4...	Windows10	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...
Mar 14, 2026 @ 09:25:26.4...	Windows10	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...
Mar 14, 2026 @ 09:25:26.4...	Windows10	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...
Mar 14, 2026 @ 09:25:26.4...	Windows10	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...

FIM: Recent events

Time ↓	Path	Action	Rule description	Rule Lev...	Rule Id
Mar 14, 2026 @ 09:25:30.957	c:\users\user\downloads\test\test...	modified	Integrity checksum changed.	7	550
Mar 14, 2026 @ 09:25:27.506	c:\users\user\downloads\test\test...	added	File added to the system.	5	554
Mar 14, 2026 @ 09:25:27.173	HKEY_LOCAL_MACHINE\System\C...	modified	Registry Key Integrity Checksum Ch...	5	594
Mar 14, 2026 @ 09:25:27.122	HKEY_LOCAL_MACHINE\System\C...	modified	Registry Key Integrity Checksum Ch...	5	594
Mar 14, 2026 @ 09:25:27.008	HKEY_LOCAL_MACHINE\System\C...	modified	Registry Value Integrity Checksum ...	5	750

2) Custom Detection Rules:

Overview: Custom Wazuh rules allow security analysts to create specific detection logic tailored to their environment. These rules analyze incoming logs and trigger alerts when defined conditions or suspicious patterns are detected.

Implementation:

- Created custom rules in local_rules.xml on the Wazuh Manager.

```
wazuhserver@wazuhserver-VirtualBox: ~
wazuhserver@wazuhserver-VirtualBox:~$ sudo nano /var/ossec/etc/rules/local_rules
.xml
[sudo] password for wazuhserver:
wazuhserver@wazuhserver-VirtualBox:~$
```

- Implemented rule to detect **new user account creation**.

```
<rule id="100101" level="10">
  <if_sid>5501</if_sid> <!-- base Wazuh user creation rule -->
  <description>New Linux user account created</description>
  <mitre>
    <id>T1136</id>
  </mitre>
</rule>

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

- Creating new user on Linux.

```
┌──(kali㉿kali)-[~]
└─$ sudo useradd testuser2

┌──(kali㉿kali)-[~]
└─$
```

- The custom rule successfully generated an alert when a new user account was created on the Linux system using the **useradd** command. The event appeared in the Wazuh dashboard with the assigned rule ID and severity level, confirming that the rule correctly detected account creation activity.

295 hits

Mar 13, 2026 @ 12:03:34.009 - Mar 14, 2026 @ 12:03:34.009

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timestamp	agent.name	rule.description	rule.level	rule.id
Mar 14, 2026 @ 12:03:27.7...	KaliLinux	New group added to the ...	8	5901
Mar 14, 2026 @ 12:03:27.7...	KaliLinux	PAM: Login session closed.	3	5502
Mar 14, 2026 @ 12:03:27.7...	KaliLinux	PAM: Login session open...	3	5501
Mar 14, 2026 @ 12:03:27.7...	KaliLinux	New user account create...	10	100004

New user account created on Linux host (Custom Rule)

View alerts of this Rule

Information

ID	Level	File	Path
100004	10	local_rules.xml	etc/rules

Groups
account_management, linux, creation, local, local, syslog, sshd

Details

If_sid
5902

Compliance

MITRE Techniques	MITRE Tactics
Create Account	Persistence

- Implemented rule to detect **SSH brute force attempts**.

```
<!-- Rule 2: SSH Brute Force Detection -->
<rule id="100002" level="12" frequency="6" timeframe="120">
    <if_matched_sid>5710</if_matched_sid>
    <same_source_ip/>
    <description>SSH brute force attack detected: >5 failures from same IP in 2
min</description>
    <mitre><id>T1110</id></mitre>
    <group>authentication_failure,brute_force,ssh</group>
</rule>
```

- Attempted brute force on linux machine

```
┌──(kali㉿kali)-[~]
└─$ ssh wronguser@[redacted]
wronguser@[redacted]'s password:
Permission denied, please try again.
wronguser@[redacted]'s password:
Permission denied, please try again.
wronguser@[redacted]'s password:
wronguser@[redacted]: Permission denied (publickey,password).

┌──(kali㉿kali)-[~]
└─$ ssh wronguser@[redacted]
wronguser@[redacted]'s password:
55555555
Permission denied, please try again.
wronguser@[redacted]'s password:
Permission denied, please try again.
wronguser@[redacted]'s password:
wronguser@[redacted]: Permission denied (publickey,password).

┌──(kali㉿kali)-[~]
└─$ ssh wronguser@[redacted]
wronguser@[redacted]'s password:
Permission denied, please try again.
wronguser@[redacted]'s password:
Permission denied, please try again.
wronguser@[redacted]'s password:
wronguser@[redacted]: Permission denied (publickey,password).

┌──(kali㉿kali)-[~]
└─$ ssh wronguser@[redacted]
wronguser@[redacted]'s password:
Permission denied, please try again.
wronguser@[redacted]'s password:
Permission denied, please try again.
wronguser@[redacted]'s password:
wronguser@[redacted]: Permission denied (publickey,password).
```

- During testing, multiple failed SSH login attempts were generated from the attacker machine. After exceeding the configured threshold, the custom rule triggered an alert in the Wazuh dashboard, successfully identifying the activity as a potential SSH brute force attack.

272 hits

Mar 13, 2026 @ 11:50:25.752 - Mar 14, 2026 @ 11:50:25.752

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timestamp	agent.name	rule.description	rule.level	rule.id
Mar 14, 2026 @ 11:41:29.3...	KaliLinux	syslog: User missed the ...	10	2502
Mar 14, 2026 @ 11:41:27.3...	KaliLinux	SSH brute force attack d...	12	100002
Mar 14, 2026 @ 11:41:23.3...	KaliLinux	sshd: Attempt to login usi...	5	5710
Mar 14, 2026 @ 11:41:21.2...	KaliLinux	sshd: Attempt to login usi...	5	5710
Mar 14, 2026 @ 11:41:19.2...	KaliLinux	PAM: User login failed.	5	5503
Mar 14, 2026 @ 11:41:19.2...	KaliLinux	sshd: Attempt to login usi...	5	5710
Mar 14, 2026 @ 11:41:17.2...	KaliLinux	syslog: User missed the ...	10	2502
Mar 14, 2026 @ 11:41:15.2...	KaliLinux	sshd: Attempt to login usi...	5	5710
Mar 14, 2026 @ 11:41:13.2...	KaliLinux	sshd: Attempt to login usi...	5	5710
Mar 14, 2026 @ 11:41:09.2...	KaliLinux	SSH brute force attack d...	12	100002
Mar 14, 2026 @ 11:41:07.2...	KaliLinux	PAM: User login failed.	5	5503
Mar 14, 2026 @ 11:41:05.2...	KaliLinux	sshd: Attempt to login usi...	5	5710
Mar 14, 2026 @ 11:41:03.2...	KaliLinux	sshd: Attempt to login usi...	5	5710
Mar 14, 2026 @ 11:41:01.2...	KaliLinux	sshd: Attempt to login usi...	5	5710
Mar 14, 2026 @ 11:40:55.1...	KaliLinux	sshd: Attempt to login usi...	5	5710

SSH brute force attack detected: >5 failures from same IP in 2 min

View alerts of this Rule

Information

ID	Level	File	Path
100002	12	local_rules.xml	etc/rules

Groups
authentication_failure, brute_force, ssh, local, syslog, sshd

Details

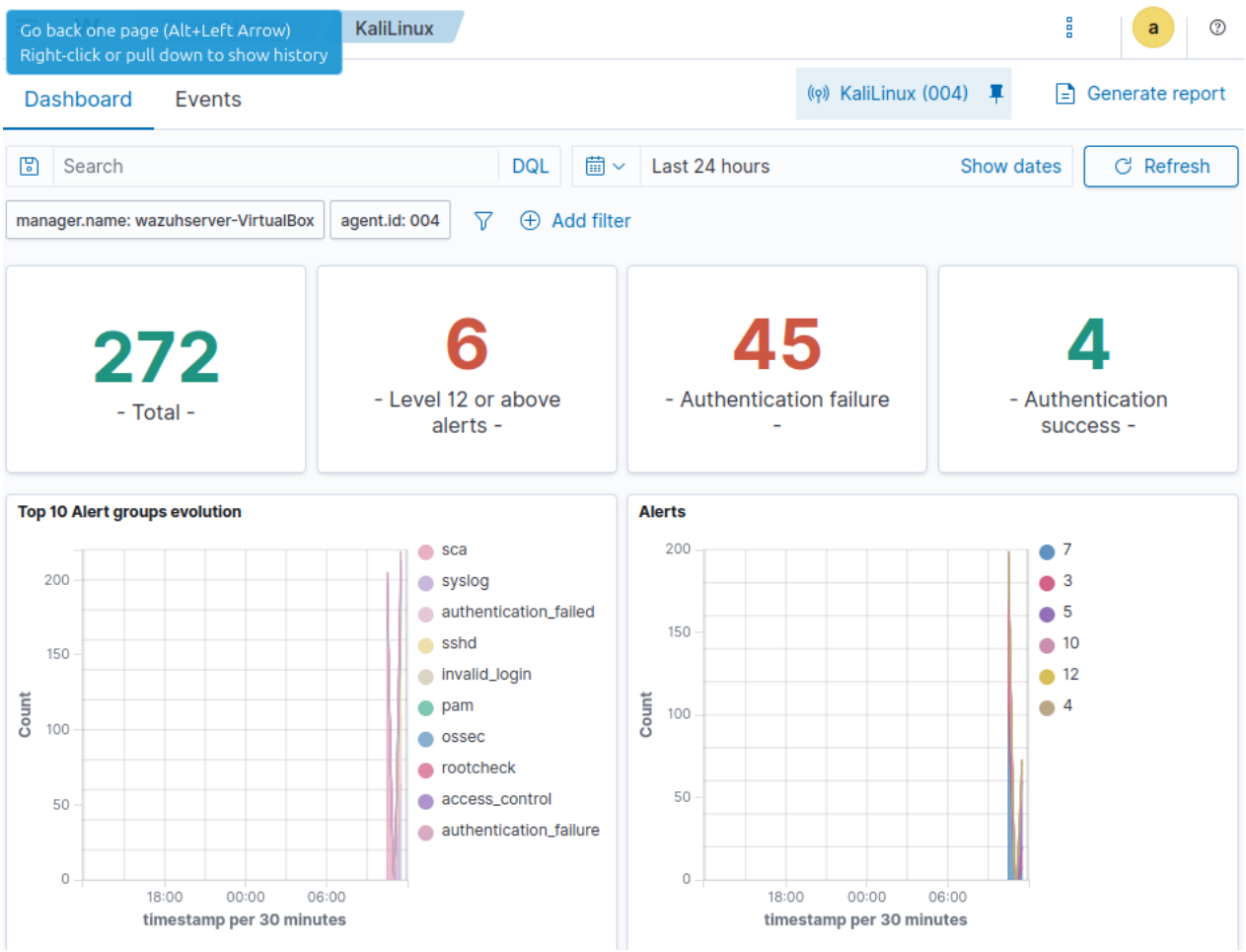
Frequency	Timeframe	If_matched_sid	Same_source_ip
6	120	5710	true

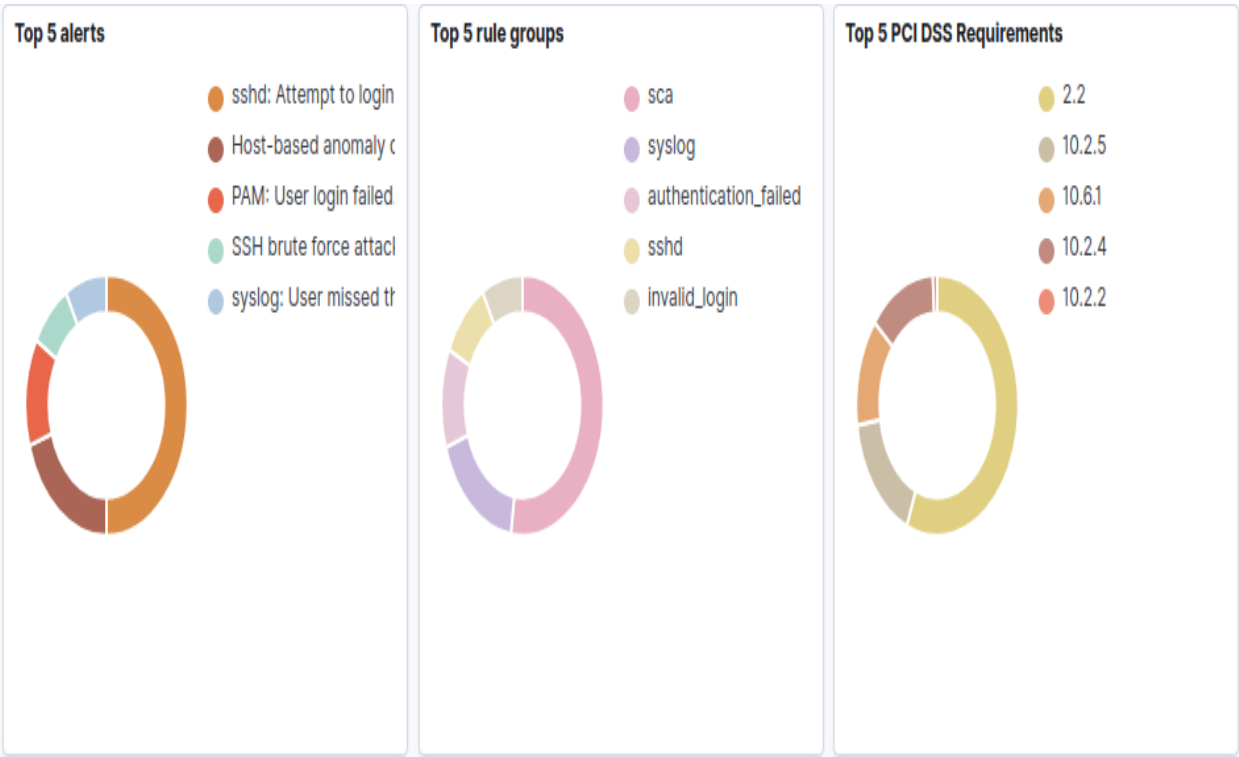
Compliance

MITRE Techniques	MITRE Tactics
Brute Force	Credential Access

Related rules

Description	Groups	Compliance	L...	File
	authen			


Go back one page (Alt+Left Arrow)
Right-click or pull down to show history
KaliLinux
Dashboard
Events
KaliLinux (004)
Generate report
Search
DQL
Last 24 hours
Show dates
Refresh
manager.name: wazuhserver-VirtualBox
agent.id: 004
Add filter
272
- Total -
6
- Level 12 or above alerts -
45
- Authentication failure -
4
- Authentication success -
Top 10 Alert groups evolution
Count
timestamp per 30 minutes
sca
syslog
authentication_failed
sshd
invalid_login
pam
ossec
rootcheck
access_control
authentication_failure
Alerts
Count
timestamp per 30 minutes
7
3
5
10
12
4


Top 5 alerts
sshd: Attempt to login
Host-based anomaly c
PAM: User login failed
SSH brute force attacl
syslog: User missed th
Top 5 rule groups
sca
syslog
authentication_failed
sshd
invalid_login
Top 5 PCI DSS Requirements
2.2
10.2.5
10.6.1
10.2.4
10.2.2

3) Active Response Automation:

Overview: Active Response in Wazuh allows the system to automatically take defensive actions when a specific alert is triggered. Instead of only generating alerts, Wazuh can execute predefined commands to block or contain suspicious activity.

Implementation:

- Configured an **active response rule** for the SSH brute force detection alert.

```
wazuhserver@wazuhserver-VirtualBox: ~
GNU nano 7.2                    /var/ossec/etc/ossec.conf

  <active-response>
    <command>firewall-drop</command>
    <location>any</location>
    <rules_id>100002</rules_id>   <!-- SSH brute force custom rule -->
    <timeout>300</timeout>
    active-response options here
  </active-response>
```

- Brute Force was attempted.

```
┌──(kali㉿kali)-[~]
└─$ ssh wronguser@[redacted]
wronguser@[redacted]'s password:
Permission denied, please try again.
wronguser@[redacted]'s password:
Permission denied, please try again.
wronguser@[redacted]'s password:
wronguser@[redacted]: Permission denied (publickey,password).

┌──(kali㉿kali)-[~]
└─$ ssh wronguser@[redacted]
wronguser@[redacted]'s password:
1
```

- When the brute force rule was triggered, Wazuh automatically executed the response to block the attacking IP address.

```
wazuhserver@wazuhserver-VirtualBox: ~
arameters.alert.id: 1773492097.4801781
arameters.alert.previous_output: Mar 14 12:41:42 kali sshd-session[5824]: Inval
d user wronguser from [redacted] port [redacted]
ar 14 12:41:37 kali sshd-session[5708]: Failed password for invalid user wrongu
er from [redacted] port [redacted] ssh2
ar 14 12:41:35 kali sshd-session[5708]: Failed password for invalid user wrongu
er from [redacted] port [redacted] ssh2
ar 14 12:41:31 kali sshd-session[5708]: Failed password for invalid user wrongu
er from [redacted] port [redacted] ssh2
ar 14 12:41:27 kali sshd-session[5708]: Invalid user wronguser from [redacted].
 port [redacted]
arameters.alert.full_log: Mar 14 12:41:45 kali sshd-session[5824]: Failed passw
rd for invalid user wronguser from [redacted] port [redacted] ssh2
arameters.alert.predecoder.program_name: sshd-session
arameters.alert.predecoder.timestamp: Mar 14 12:41:45
arameters.alert.predecoder.hostname: kali
arameters.alert.decoder.parent: sshd
arameters.alert.decoder.name: sshd
arameters.alert.data.srcip: [redacted]
arameters.alert.data.srcuser: wronguser
arameters.alert.location: journald
arameters.program: active-response/bin/firewall-drop
```

- During testing, repeated SSH login attempts were generated from the Kali machine. Once the rule threshold was reached, Wazuh triggered the active response and automatically added a firewall rule blocking the attacker IP, which was confirmed using the iptables -L -n command.

```
┌──(kali㉿kali)-[~]
└─$ sudo iptables -L -n
[sudo] password for kali:
Chain INPUT (policy ACCEPT)
target     prot opt source               destination
DROP       all  --  [redacted]           0.0.0.0/0

Chain FORWARD (policy ACCEPT)
target     prot opt source               destination
DROP       all  --  [redacted]           0.0.0.0/0

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination

┌──(kali㉿kali)-[~]
└─$
```

Conclusion:

In this lab, the existing SOC home lab environment was further expanded by implementing additional security monitoring and response capabilities using Wazuh. File Integrity Monitoring was configured to detect changes in sensitive system directories, while custom detection rules were created to identify events such as new user creation and SSH brute force attempts.

Active Response was also implemented to automatically block malicious activity. During testing, repeated SSH login attempts triggered the configured rule and Wazuh successfully blocked the attacking IP using firewall rules. Overall, this lab demonstrated how detection, monitoring, and automated response can work together in a SOC environment to identify and mitigate potential security threats.